Assignment - Cloud

Course: 696a941b009abdace012e41a | Due Date: 1/23/2026
Questions: 5 | Total Marks: 50
Type: Descriptive Questions

QUESTIONS:

Q1. Explain the concept of cloud computing and its key characteristics. Provide examples of popular cloud service providers.

[10 Marks | Difficulty: medium]

Expected Points:

- Definition of cloud computing
- Explanation of key characteristics (e.g., on-demand self-service, broad network access, resource pooling, rapid elasticity, measured service)
- Examples of popular cloud service providers (e.g., Amazon Web Services, Microsoft Azure, Google Cloud Platform)

Q2. Discuss the differences between public, private, and hybrid clouds. Provide advantages and disadvantages of each type.

[10 Marks | Difficulty: medium]

Expected Points:

- Explanation of public, private, and hybrid clouds
- Advantages and disadvantages of public cloud
- Advantages and disadvantages of private cloud
- Advantages and disadvantages of hybrid cloud

Q3. Describe the essential components of a cloud computing architecture. Explain the roles of virtualization, networking, and storage in cloud environments.

[10 Marks | Difficulty: medium]

Expected Points:

- Explanation of essential components of cloud computing architecture (e.g., front-end, back-end, cloud storage)
- Role of virtualization in cloud computing
- Role of networking in cloud computing
- Role of storage in cloud computing

Q4. Examine the security challenges in cloud computing. Discuss common threats and measures to enhance cloud security.

[10 Marks | Difficulty: medium]

Expected Points:

- Identification of security challenges in cloud computing (e.g., data breaches, data loss, insecure APIs)
- Explanation of common threats in cloud computing
- Measures to enhance cloud security (e.g., encryption, access control, regular audits)

Q5. Explore the concept of serverless computing in the cloud. Compare serverless computing with traditional server-based models. Provide use cases for serverless computing.

[10 Marks | Difficulty: medium]

Expected Points:

- Explanation of serverless computing and its key features
- Comparison between serverless computing and traditional server-based models
- Use cases for serverless computing (e.g., event-driven applications, real-time file processing)